

GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)



DEPARTMENT OF MECHANICAL ENGINEERING

(2025-2026)

A PROJECT ON

**“DESIGN AND FABRICATION OF PARABOLIC WATER HEATER
WITH SOLAR TRACKING AND POWER GENERATION”**

UNDER THE GUIDANCE OF

Prof. A. R. BANSALI

~SUBMITTED BY~

MS. ASHWINI R. HAGE (23ME031)

MR. ROHAN S. HAGE (23ME032)

MR. AMAY P. KADAM (23ME044)

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~SUBMITTED TO~

GOVERNMENT POLYTECHNIC, AMRAVATI

IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF

DIPLOMA IN MECHANICAL ENGINEERING

~SUBMITTED BY~

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LECTURER IN MECHANICAL ENGINEERING

GOVERNMENT POLYTECHNIC AMRAVATI

(2025-2026)

GOVERNMENT POLYTECHNIC, AMRAVATI

(An Autonomous Institute of Government of Maharashtra)
Department of Mechanical Engineering



Certificate

This is to certify that, **Mr. Amay P. Kadam (23ME044)** of 3rd year (Odd term) Diploma in Mechanical Engineering have satisfactorily completed the major project entitled “**DESIGN AND FABRICATION OF PARABOLIC WATER HEATER WITH SOLAR TRACKING AND POWER GENERATION**” of course, **ME7501- Capstone Project** for the academic year 2025-2026 as prescribed in curriculum.

Prof. A. R. Bansali

Project Guide
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Dr. A. B. Borade

(Principal)
(Government polytechnic, Amravati)

GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

VISION

- To be a vibrant technical institute of global repute contributing towards the needs of industries & society.

MISSION

- To develop competent diploma engineers suitable for contemporary industrial environment.
- To inculcate socially accepted ethics & values among budding engineers.
- To Nurture innovations and entrepreneurship.
- To produce engineers with psychomotor & cognitive skills committed to lifelong learning.

MECHANICAL ENGINEERING DEPARTMENT

VISION

- To become center of excellence in Mechanical Engineering, catering to the needs of industry and society.

MISSION

- Provide excellent academic ambience for imparting knowledge and skills.
- Inculcate awareness about behavioral skills, ethical practices and sustainable development.
- Provide nurturing environment for promoting entrepreneurship.
- Respond effectively to the needs of the industry and changing world.

MECHANICAL ENGINEERING DEPARTMENT

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1- Technical competence: Diploma holders will be successful in modern mechanical engineering practice.

PEO2- Professional Skills: Diploma holders will continue to acquire and demonstrate the professional skills necessary to be competent employees or entrepreneurs.

PEO3- Professional Attitude and Citizenship: Diploma holders will become productive and responsible citizens with high ethical and professional standards.

PROGRAMME OUTCOMES (POs)

PO1: Basic and Discipline Specific Knowledge: Apply knowledge of basic mathematics, science and engineering fundamentals and engineering specialization to solve the engineering problems.

PO2: Problem analysis: Identify and analyze well defined engineering problems using codified standard methods.

PO3: Design/development solutions: Design solutions for well defined technical problems and assists with the design of systems components or processes to meet specified needs.

PO4: Engineering tools, Experimentation and testing: Apply modern engineering tools and appropriate technique to conduct standard tests and measurements.

PO5: Engineering practices for Society, sustainability and environment: Apply appropriate technology in context of society, sustainability, environment and ethical practices.

PO6: Project Management: Use engineering management principles individually, as team member or a leader to manage projects effectively communicate about well-defined engineering activities.

PO7: Life-long learning: Ability to analyze individual needs and engage in updating in the context of technological changes.

PROGRAM SPECIFIC OUTCOMES (PSOs):

PSO1: To inculcate and foster entrepreneurship.

PSO2: To plan, design, fabricate, test, operate and maintain simple mechanical systems.

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Department of Mechanical Engineering



Declaration

I am the student of Mechanical Engineering (**Section-A**) of **Government Polytechnic, Amravati** and I hereby declare that the entire work embodied in this Major-Project Report Entitled "**DESIGN AND FABRICATION OF PARABOLIC STEAM GENERATION WITH SOLAR TRACKING AND POWER GENERATION**" has been carried out by me under the supervision and guidance of Shri. A. R. Bansali, Project Guide & Lecturer in Mechanical Engineering. I further declare that no part of this report has not been previously submitted for any degree, diploma or to any other diploma examining Body or University.

Place: - Amravati,

Date: - / /2025.

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This would not have been possible without guidance and help of many people. This is only page where I have opportunity of expressing my emotions and gratitude from the bottom of my heart to them.

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I would like to thank **Dr. A. B. Borade** Principal of our institute for providing necessary facilities during the period of working on this project.

Last but not the least; I would like to express our thankfulness to teaching and non-teaching staff, my friends and well-wishes.

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ABSTRACT

The increasing global energy demand and depletion of fossil fuels have accelerated the need for sustainable energy solutions. Solar energy, being clean, renewable, and abundant, is a reliable alternative for thermal and electrical applications. In this work, a solar water heater and steam generator system is designed and fabricated using a cylindrical parabolic concentrator (CPC). The CPC focuses incident solar radiation onto a receiver tube, thereby enhancing the thermal efficiency. Unlike flat plate collectors, the concentrator achieves higher temperatures suitable for both water heating and steam generation. The proposed design ensures optimum utilization of solar energy for domestic, industrial, and agricultural applications.

The system incorporates real-time automated solar tracking to overcome limitations of fixed-angle collectors. A dual-axis tracking mechanism powered by sensors and microcontroller logic adjusts the concentrator to maintain perpendicularity to solar rays. This ensures maximum concentration of energy throughout the day, significantly improving thermal output. The tracking system reduces losses caused by diurnal variations and seasonal shifts in solar position. Automation also minimizes human intervention and enhances operational reliability. With improved efficiency, the system provides consistent steam production even during partial sunlight conditions.

The fabricated prototype consists of a cylindrical concentrator, absorber tube, insulated water storage, and control electronics. Mild steel and reflective aluminum are used for the concentrator frame and surface to ensure durability and high reflectivity. The receiver tube, coated with selective absorptive material, minimizes heat losses while maximizing absorption. Heat transfer to the working fluid (water) occurs rapidly, raising its temperature to produce hot water and steam. Insulated storage prevents thermal losses and allows continuous supply even after sunset. The compact and modular design enables easy scalability for domestic as well as industrial usage.

Experimental testing demonstrated the system's ability to achieve water heating up to boiling point and generate steam at controlled rates. Real-time tracking improved efficiency by 30–40% compared to fixed concentrators. The study concludes that the developed system is an efficient, low-cost, and eco-friendly alternative to conventional water heating and steam generation methods. It offers significant potential in rural areas where grid energy is limited. Additionally, the system reduces carbon emissions, promoting green technology adoption. Further research may involve hybridization with other renewable systems and performance optimization under varying climatic conditions.



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CHAPTER-1

INTRODUCTION

1.1 OVERVIEW

Energy plays a vital role in the development of any country. The increasing demand for energy and the depletion of conventional energy sources like coal, oil, and natural gas have made it necessary to explore renewable energy resources. Among these, solar energy is the most abundant, clean, and sustainable form of energy available.

The Sun radiates a huge amount of energy in the form of light and heat. If effectively utilized, this energy can fulfill a major part of the world's energy needs. Solar energy can be converted into useful heat energy or electrical energy through various technologies such as solar water heaters, photovoltaic cells, and solar cookers.

The present project, titled “Design and Fabrication of Parabolic Water Heater with Solar Tracking and Power Generation,” is a step toward utilizing this renewable energy for practical use. The system focuses on capturing solar radiation using a parabolic collector to heat water efficiently and uses solar tracking to improve performance. It also integrates a small-scale power generation unit, making it a multipurpose and eco-friendly system.

1.2 NEED OF PROJECT

In most rural and semi-urban areas, people still depend on firewood, kerosene, and LPG gas for heating water. These fuels not only increase household expenses but also contribute to pollution and environmental degradation.

Using solar energy for heating water is an ideal solution because:

It is freely available and inexhaustible.

It reduces dependency on non-renewable energy sources.

It helps save fuel and electricity.

It lowers carbon emissions and environmental pollution.

However, most existing solar water heaters are fixed in one direction and cannot follow the Sun's movement throughout the day. This results in a considerable loss of efficiency. To overcome this problem, a solar tracking mechanism is introduced to maintain the collector's orientation toward the Sun, thereby increasing heat gain and system output.

1.3 OBJECTIVE

1. To design and develop a parabolic concentrator capable of focusing solar rays onto a focal point for efficient heat generation.
2. To fabricate a cost-effective parabolic water heater using locally available materials.
3. To utilize solar energy for heating water and producing steam suitable for domestic or small-scale industrial applications.
4. To design and implement an automatic solar tracking system for continuous alignment of the parabola with the sun.
5. To enhance the thermal efficiency of the system by maintaining maximum solar exposure throughout the day.
6. To integrate a solar panel for generating electrical power required to operate the tracking motor and control system.
7. To create a self-sustaining hybrid system capable of both thermal and electrical energy generation.
8. To analyze the performance of the parabolic heater under different climatic and sunlight conditions.
9. To reduce dependency on non-renewable energy sources through the use of clean solar energy.
10. To promote the application of renewable technologies for domestic water heating and rural energy solutions.
11. To evaluate the temperature rise, efficiency, and overall performance of the fabricated system.
12. To demonstrate the environmental benefits of using solar-based hybrid energy system

CHAPTER-2

Literature Review

Literature Survey Solar energy has been harnessed for heating and power generation for several centuries, dating back to ancient civilizations that used mirrors and lenses to concentrate sunlight for practical applications. Early research on solar concentrators and parabolic reflectors in the 18th and 19th centuries laid the foundation for modern solar thermal systems. Initial designs were simple, often stationary reflectors used for heating water or cooking food, and lacked automation or efficiency optimization, but they demonstrated the potential of concentrated solar energy.

In the latter half of the 20th century, advancements in materials, reflectors, and control systems enabled the development of more efficient parabolic troughs and dishes. Researchers focused on improving the concentration ratio, heat transfer efficiency, and receiver design. During this period, solar tracking mechanisms also began to emerge, initially as manual or semi-automatic systems, allowing solar concentrators to follow the sun's path and improve energy capture. These developments highlighted the importance of combining solar tracking with concentrators for optimal performance.

In the present era, significant improvements in photovoltaic (PV) technology and automated solar tracking systems have made hybrid systems combining thermal and electrical energy feasible. Modern parabolic water heaters now integrate solar tracking, temperature sensors, and electric motors powered by solar panels, creating partially self-sustaining systems. Studies in recent years have shown that such hybrid systems can improve overall energy utilization by up to 30–50% compared to fixed or non-tracked systems. Additionally, the use of readily available materials and cost-effective fabrication methods has made these technologies accessible for both domestic and industrial applications.

Future trends in parabolic water heating and power generation focus on smart automation, AI-based tracking, and multi-functional hybrid systems. Researchers are exploring the integration of IoT devices for real-time monitoring, predictive control of solar tracking, and energy storage optimization. Advanced materials like selective coatings for receivers and high-reflectivity mirrors are also being developed to enhance heat absorption and reduce energy losses. These innovations aim to create more efficient, durable, and environmentally friendly solar systems capable of meeting increasing energy demands.

In summary, the evolution of parabolic water heaters reflects a continuous effort to maximize the efficiency of solar energy utilization. From simple stationary concentrators in the past to modern automated hybrid systems today, and towards intelligent, smart solar devices in the future, research consistently aims to improve thermal and electrical output, reduce costs, and promote sustainable energy solutions. This project builds on this foundation by designing and fabricating a parabolic water heater with solar tracking and power generation, contributing to the growing field of renewable energy technology.

This Table Summarizes The Literature Survey

Aspect	Present
Technology Type	Automated parabolic dishes with solar tracking; hybrid thermal and PV systems
Energy Utilization	Thermal + electrical energy (heating + power generation)
Efficiency	Moderate to high (30–50% improvement with tracking and hybrid integration)
Materials	Improved reflectors, insulated receivers, durable metals, and solar panels
Automation	Motorized solar tracking, sensor-based alignment
Environmental Impact	Clean energy with partial reduction of fossil fuel use
Cost	Moderate cost with improved efficiency
Applications	Domestic and small industrial heating, electricity generation Smart homes

CHAPTER-3

METHODOLOGY

3.1 DESING PARAMETER AND SELECTION OF COMPONENTS

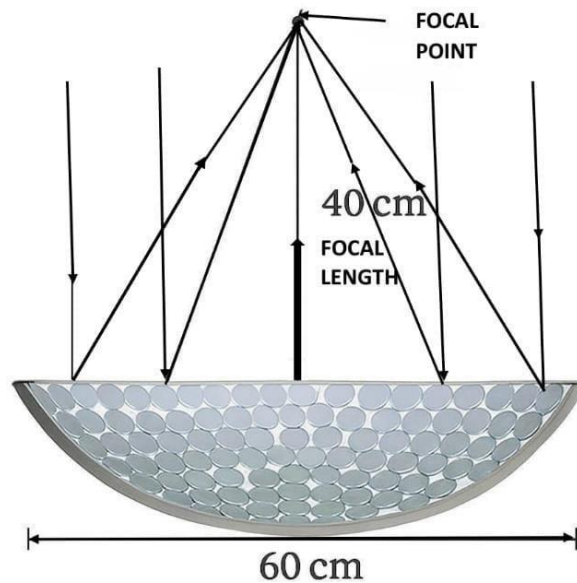
1) PARABOLIC DISH



The image shows the fabricated parabolic solar concentrator developed for this project, which is designed to focus sunlight onto a single focal point for efficient heat generation. The concentrator surface is constructed using multiple small square mirror pieces carefully arranged on a curved parabolic base to achieve the desired reflective geometry. Each mirror segment reflects incoming solar radiation toward the focal point, where a receiver tube or container is placed to absorb concentrated heat energy. This design allows the system to attain high temperatures suitable for water heating and steam generation, utilizing clean and renewable solar energy.

The use of mirror tiles instead of a single reflective sheet significantly reduces fabrication costs while maintaining high reflectivity and durability. The circular structure provides better light focusing and structural stability. The concentrator, with a 60 cm diameter and 40 cm focal length, is integrated with a real-time automated solar tracking system to maintain optimal alignment with the sun throughout the day. This practical and cost-effective design demonstrates the potential of locally fabricated solar concentrators for sustainable thermal energy applications in domestic and industrial sectors. A parabolic concentrator is a reflective device that focuses incoming solar radiation onto a focal point or line. The design is based on the geometric properties of a parabola, where all rays parallel to the principal axis are reflected through a single focal point. In this project, a cylindrical parabolic concentrator is used to concentrate sunlight onto a receiver tube located at the focal point to achieve efficient heating and steam generation.

DESIGN PARAMETERS



From the given design:

Aperture Diameter (D): 60 cm

Focal Length (f): 40 cm

Shape: Cylindrical Parabolic Reflector

Type: Linear-focus concentrator

Material: mirror-finished sheet

Receiver Position: Placed along the focal line at a height equal to the focal length (40 cm from the vertex)

2) RECEIVER



In this project, the receiver plays a vital role in absorbing concentrated solar radiation and converting it into thermal energy. The receiver is fabricated from a copper coil with an internal diameter of 3 mm, selected for its excellent thermal conductivity and corrosion resistance. Copper is an ideal material for solar thermal systems as it ensures rapid heat transfer from the concentrated sunlight to the working fluid circulating inside the coil.

The receiver coil is positioned precisely at the focal point of the parabolic concentrator, where maximum solar energy is focused.

To enhance heat absorption and minimize reflective losses, the copper coil is coated with matte black heat-resistant paint. The black surface increases the absorptivity of the receiver, allowing it to capture a greater portion of incident solar radiation. The small internal diameter ensures a higher rate of heat transfer to the flowing water, enabling quick temperature rise and even steam generation under continuous solar exposure. This receiver design ensures efficient energy utilization and contributes significantly to the overall performance and thermal efficiency of the solar water heater and steam generator system.

3) COLD WATER STORAGE TANK (Pre-Heating Unit)



A black-colored cold storage water tank is installed as a pre-heating unit in the system. It serves two purposes:

SPECIFICATION:

1. Pre-heating of water: The black surface absorbs solar radiation effectively, slightly increasing the temperature of the water before it enters the copper receiver. This pre-heating process improves overall system efficiency and reduces the time required to reach the desired hot-water temperature.
2. Storage: The tank also acts as a temporary storage reservoir that supplies a steady flow of water to the parabolic collector.

The tank is made of lightweight aluminium material, coated with black paint to maximize solar absorption.

A small outlet at the bottom connects to the copper receiver tube through a flexible pipe. This simple addition improves energy utilization and makes the design cost-effective and efficient.

3. Capacity of the tank : 5 lit

4) WATER SUPPLY PIPE (Connecting Line)



A flexible blue plastic pipe is used in the system to connect the cold storage water tank with the copper receiver tube.

This pipe serves as the main water supply line and ensures a smooth, continuous flow of water from the tank to the receiver during operation.

Material and Function:

The pipe is made of high-density polyethylene (HDPE), which is lightweight, durable, and resistant to heat and corrosion.

It is capable of withstanding moderate pressure and temperature, making it suitable for small scale solar heating systems.

The inner surface is smooth, allowing easy water flow with minimal frictional losses.

Its blue color helps in identifying it as the cold-water line, preventing confusion with the hot-water outlet.

Role in the Project:

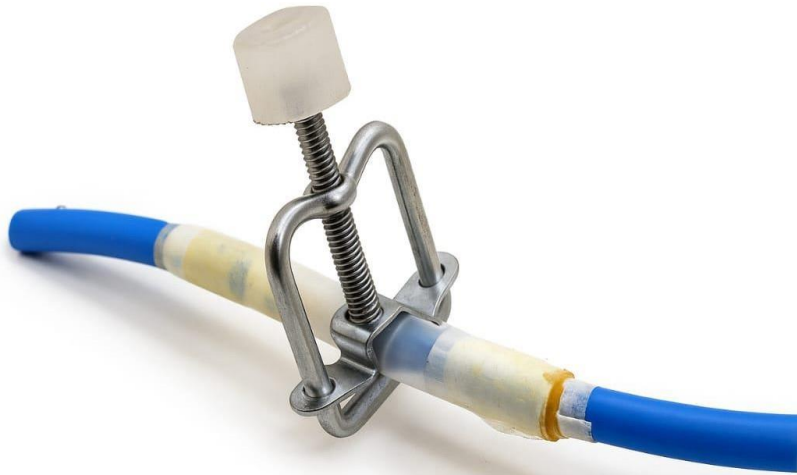
Carries water from the pre-heating storage tank to the copper receiver.

Maintains a constant water level and flow rate in the heating section.

Improves system performance by ensuring that no air pockets or leaks occur.

Flexible enough to accommodate small movements in the solar tracking assembly.

5) FLOW CONTROL VALVE (Hoffman Clamp Arrangement)



A Hoffman clamp is used in the project as a manual flow control valve for the water passing through the blue supply pipe.

It allows precise regulation of the water flow from the storage tank to the copper receiver tube, ensuring that the heating process remains efficient and steady.

Construction:

The clamp consists of a U-shaped metal frame with a threaded screw and a knob at the top. The blue flexible pipe passes through the clamp opening.

By rotating the knob clockwise, the screw presses against the pipe, restricting or stopping the water flow.

Turning it anti-clockwise loosens the pressure, allowing smooth water passage.

Function in the Project:

1. Controls the rate of water flow to maintain proper heat absorption in the copper tube.
2. Prevents wastage of hot water and maintains uniform temperature rise.
3. Acts as a simple and low-cost valve system, eliminating the need for expensive plumbing fittings.
4. Makes the experimental setup easy to adjust during testing. This component ensures better temperature control and improves the overall performance and efficiency of the solar water heater.

6) ARDUINO NANO



The Arduino Nano is a compact, microcontroller-based development board used as the main control unit in the solar tracking system of this project. It is responsible for processing the signals received from the Light Dependent Resistors (LDRs) and controlling the PMDC motor through the motor driver module, ensuring that the parabolic collector continuously follows the Sun's position throughout the day.

Function:

Two LDR sensors are mounted on opposite sides of the collector.

When sunlight intensity differs between the two sensors, the Arduino Nano detects the voltage difference.

It sends a control signal to the motor driver circuit, which activates the PMDC motor to rotate the collector toward the brighter side (toward the Sun).

Once both LDRs sense equal light, the Arduino stops the motor — meaning the collector is correctly aligned with the Sun.

7) MOTOR DRIVER UNIT

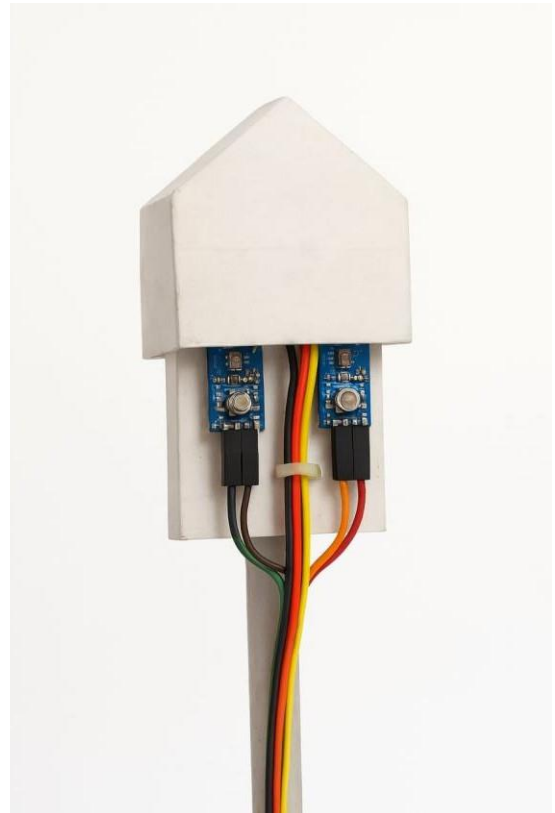
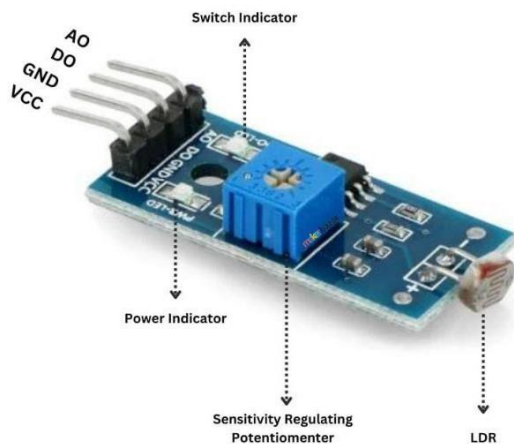


The motor driver unit used in my project is an L298N motor driver module, which is designed to control the speed and direction of a Permanent Magnet DC (PMDC) motor. It acts as an interface between the microcontroller and the motor because a microcontroller cannot supply the high current required by the motor. The L298N is a dual H-bridge motor driver IC that can control two PMDC motors or one stepper motor. It operates using an external power supply (usually 5V–35V) and provides sufficient current (up to 2A per channel) to drive the motors efficiently.

Function:

The motor driver works on the H-bridge principle, which allows the motor to rotate in both forward and reverse directions. When control signals from the microcontroller are applied to the input pins, the driver sends the proper polarity of voltage to the motor terminals. By changing the logic of the inputs, the direction of current through the motor changes, which changes the rotation direction. The enable pins control the motor speed using Pulse Width Modulation (PWM) signals. Thus, the L298N motor driver helps to easily control the speed and direction of a PMDC motor in the project.

8) LDR SENSOR MODULE (Light Detection Unit)



The LDR (Light Dependent Resistor) sensor module is used in this project to detect the intensity of sunlight and provide input to the Arduino Nano for automatic solar tracking. It works on the principle that the resistance of an LDR changes with the intensity of light falling on it.

Function:

Two LDR modules are placed at slightly different positions on the parabolic collector. When sunlight intensity is unequal between the two sensors, the resistance difference creates a voltage difference this voltage difference is sent to the Arduino Nano, which processes the signal. The Arduino then commands the motor driver to rotate the collector toward the side receiving less light, aligning it perfectly with the Sun once both LDRs sense equal sunlight, the system stops the motor — indicating correct alignment.

9) PUSH BUTTONS



Push button switches are used as control inputs to operate the motor's direction of rotation. These push buttons are simple, momentary-type switches that make or break an electrical connection when pressed. They are compact, easy to use, and reliable for controlling electronic circuits. Each button is connected to the control pins of the motor driver circuit, which in turn determines the motor's movement direction.

Function:

There are two push buttons used in the project — one for right (forward) rotation and the other for left (reverse) rotation of the PMDC motor. When the right push button is pressed, it sends a signal to the motor driver to rotate the motor in the forward direction. Similarly, pressing the left push button changes the signal polarity through the driver, causing the motor to rotate in the reverse direction. Once the button is released, the circuit opens, and the motor stops. This simple push button control allows easy and manual direction control of the motor.

9) PERMANENT MAGNET DIRECT CURRENT (PDMC) MOTOR



Specifications

Voltage: 6V or 9V DC

Speed: 100–300 RPM (depends on load)

Power: 5–15 Watts

Control: Easily reversible by changing polarity

Function

1. Tracking Motion:

The motor is used to rotate the dish (either horizontally or vertically) to follow the sun's movement throughout the day. It ensures maximum solar radiation is focused on the receiver.

2. Automatic Control:

The motor operates through a motor driver and sensor system (like LDR sensors) that detect sunlight direction.

3. Power Source:

It is powered by a battery or solar panel output, making the system self-sustainable.

10) V-GROOVE BELT AND PULLEY



Specifications of V-Groove Belt and Pulley

1. Type of drive – V-belt drive
2. Belt type – V-groove
3. Belt material – Reinforced rubber
4. Belt length – Around 600 mm to 800 mm
5. Driven pulley diameter – 25 cm (250 mm)
6. Driver pulley diameter – 5 cm to 10 cm (mounted on motor shaft)
7. Speed ratio – 1 : 2.5 to 1 : 5 (for speed reduction and torque increase)
8. Driver pulley material – Mild steel or aluminum alloy
9. Driven pulley material – Fiber
10. Center distance between shafts – 175 mm to 180 mm (adjustable)

Function

1. Power Transmission:

The V-groove belt connects the PMDC motor to the dish rotation shaft via the 25 cm pulley. It transmits the rotational motion from the motor to the parabolic dish assembly.

2. Speed Reduction and Torque Increase:

A larger pulley (25 cm) helps reduce speed and increase torque, which is ideal for slow and steady dish rotation.

3. Smooth and Silent Operation:

The V-belt drive provides smooth, shock-free motion with minimal vibration and noise.

4. Protection Against Overload:

If the dish is obstructed, the belt may slip slightly, protecting the motor from damage.

5. Easy Maintenance:

Simple to install, adjust, and replace; no lubrication required.

10) BATTERY(6V,6AMP)



The system uses a Spice VRLA Battery (6V, 6Ah) made with graphene and carbon fiber technology. It is a sealed lead-acid, maintenance-free battery that provides stable DC power for operating the motor and sensors.

Main Specifications:

Type: VRLA (Valve Regulated Lead Acid) Battery

Rated Voltage: 6V

Capacity: 6Ah

Charging Voltage: 7.2V – 7.5V (Cycle use)

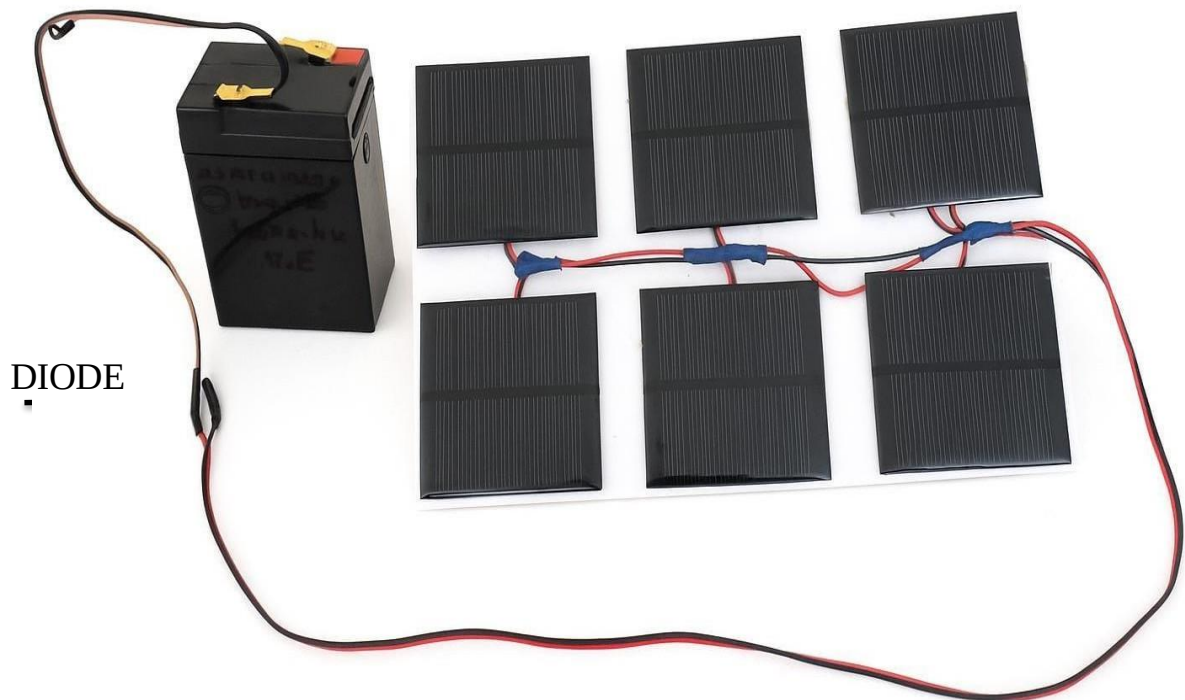
Standby Voltage: 6.78V – 6.90V

Maximum Charging Current: 1.35A

Features: Maintenance-free, long service life, and safe sealed design

This battery ensures reliable power supply for small DC motors and electronic sensor circuits in the project.

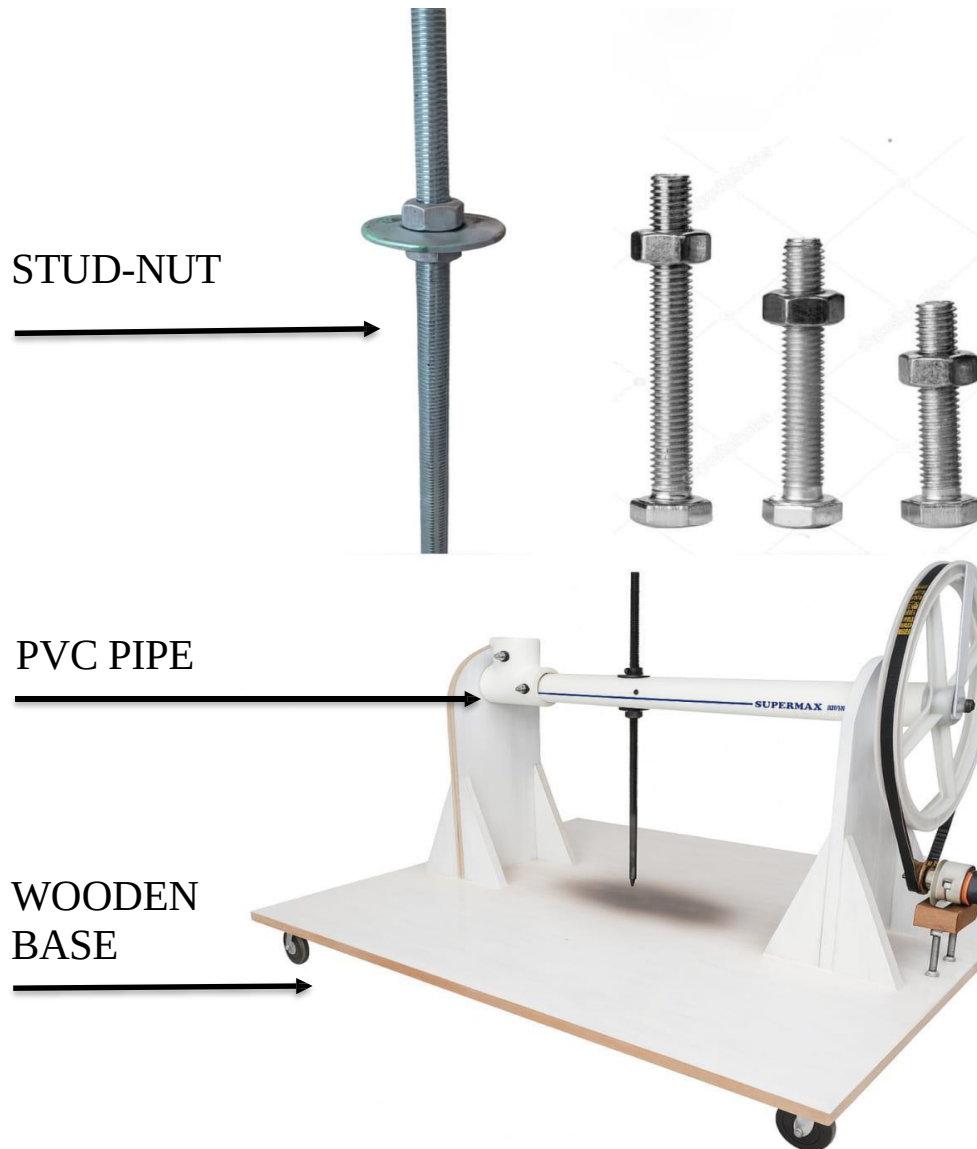
11) SOLAR PANEL UNIT



A solar panel unit is used as the main source of electrical power generation. Each solar panel is capable of producing around 6 amperes of current, and when connected together in a suitable configuration, the entire system generates a total output voltage of about 10 to 12 volts. This generated DC power is used for charging the battery, which later supplies energy to the control and motor units of the system. Solar energy is a renewable and eco-friendly source, making the setup efficient and sustainable for long-term operation.

To ensure safe operation, a diode is connected in the circuit to allow one-way power flow from the solar panels to the battery. This prevents the reverse flow of current, protecting the solar panels from short-circuiting or back current during low or no sunlight conditions. This arrangement helps maintain the reliability and safety of the charging system.

12) WOODEN BASE, PVC PIPE & VARIOUS TYPES OF NUT



The stud rod is a fully threaded rod that provides adjustable connections using nuts and washers at both ends. Different types of nuts and bolts are used to fix and support various parts of the assembly securely. The PVC pipe acts as the main shaft, on which the pulley is mounted. This shaft transmits rotational motion from the motor to the parabolic dish, allowing smooth and stable operation.

The motor and pulley arrangement are connected through a belt drive, which transfers torque efficiently to the PVC shaft. The base platform of size 60×60 mm is made from a wooden sheet, providing a strong and stable foundation for the entire setup. This design ensures that all components remain firmly in place during the operation and allows precise and balanced rotation of the parabolic dish for tracking or positioning purposes.

3.2) SOLAR TRACKING MECHANISM :



1. Introduction

The single-axis solar tracking mechanism is used to automatically rotate the parabolic dish from east to west to follow the sun's movement.

It helps maintain maximum sunlight focus on the receiver throughout the day.

The system works using LDR sensors, a PMDC motor, and a control circuit powered by solar energy.

This automatic motion increases the overall efficiency and heat output of the solar water heater. It is a simple, low-cost, and effective method for improving solar energy utilization.

○ 2. Type of Tracking

Type: Single Axis Automatic Tracking

Axis of Rotation: East–West (horizontal axis)

Purpose: To maintain continuous alignment of the parabolic dish with the moving sun during the day.

Control: Automatic using sensors and motor.

3. Components of Single Axis Tracking System

Component Specification / Function

LDR Sensors (2 No.) - Detect sunlight intensity difference on both sides

Control Circuit / Comparator - Compares LDR signals and controls motor direction

PMDC Motor (6–9V, 10–15W) - Provides rotary motion for dish movement

Motor Driver (L293D / Relay) - Drives motor forward or reverse based on sensor input

V-Groove Belt & Pulley (25 cm Pulley) - Transfers motion from motor to dish

Power Supply - 12V DC battery charged by small solar panel

Dish Support Frame Allows rotation of dish around single horizontal axis

4. Working Principle

❖ Sensing Sunlight:

- Two LDR sensors are placed on either side of a small divider.
- When sunlight falls equally on both → system is balanced → motor off.

❖ Sun Movement:

- As the sun shifts from East to West, one LDR receives more light.
- The control circuit detects the difference in voltage between LDRs.

❖ **Motor Activation:**

- The comparator or controller sends a signal to the motor driver.
- The PMDC motor rotates the dish toward the brighter side.

❖ **Balancing:**

- When both sensors receive equal light again, the motor stops.
- The dish stays aligned with the sun automatically.

❖ **End of Day:**

- The dish remains in the West position at sunset.
- It can be manually or automatically reset to the East in the morning.

5. Functions of Tracking Mechanism

- Continuously aligns the dish with the sun's position.
- Ensures maximum solar radiation is focused on the receiver.
- Eliminates need for manual adjustment.
- Increases water heating and overall system efficiency.
- Operates using solar power, making it self-sufficient.

6. Advantages

Simple and cost-effective compared to dual-axis systems.

Reduces human effort.

Compact and energy-efficient.

Ideal for small- to medium-size parabolic dish systems.

Improves output by 20–30% over fixed-position setup.

CHAPTER-4

FABRICATION

1. Frame Construction

The base frame is made of wooden sheet (60×60 cm) to provide stability and support to the setup. Holes are drilled to fix the shaft supports at the center.

The frame is painted by white colour because it reflect solar rays to prevent electronic component from over heating.

2. Parabolic Dish Fabrication

A parabolic dish is made using satellite dish for lightweight and good reflection.

The dish surface is covered with reflective mirror pieces to concentrate sunlight at the focus point. The dish is mounted on a PVC pipe shaft that allows rotational motion.

3. Receiver Fabrication

The receiver pipe (usually copper tube) is placed at the focal point of the parabola dish.

The copper tube is black-painted for maximum heat absorption. The tube is connected with inlet and outlet pipes to circulate water.

Copper tube is placed on the insulating material (aluminium plate and glass wool) for reducing heat loss.

4. Tracking Mechanism

A single-axis solar tracking system is used to follow the sun's motion east to west.

A PMDC motor with pulley and V-belt attached with the PVC pipe shaft which is used to rotate the parabolic dish. Sensors (LDRs) are placed on both sides of the dish to detect sunlight intensity difference.

A motor driver circuit and microcontroller (Arduino nano) control the motor rotation based on sensor signals.

5. Power Supplying Unit (Solar Panel & Battery)

A small solar panel (12V) is mounted on the base frame or beside the dish.

The panel converts sunlight into electrical energy to charge a 6V battery.

The battery supplies power to the motor driver, sensors, and control circuit for automatic tracking.

A blocking diode is used to ensure one-way current flow from the panel to the battery, preventing discharge at night.

6. Water Circulation System

Water from the storage tank which is placed at a certain height for direct flow of water to the inlet of receiver tube.

Heated water exits from the outlet of the receiver tube and is collected in another container.

7. Assembly & Testing

All components (frame, dish, receiver, motor, sensors, battery) are assembled.

The system is tested under sunlight to check tracking accuracy and water temperature rise.

Temperature readings are taken at regular intervals for performance analysis.

8. Finishing

Wiring is neatly arranged using cable ties.

Components are painted or coated to prevent corrosion.

Labels are added for inlet, outlet, and electrical parts for clarity.

CHAPTER-5

CONSTRUCTION



The construction of the Parabolic Solar Water Heater with Automatic Tracking and Power Generation begins with preparing a strong base structure. A wooden sheet of 60 cm × 60 cm is used as the base platform to support all components. A PVC pipe shaft is mounted horizontally at the center of this base to act as the main support for the parabolic dish. The end caps are fitted at the end of the shaft to allow smooth rotational movement. The base and supporting parts are properly aligned, drilled, and painted with white colour to prevent damage of electronic components from sunlight. All joints are fixed using nuts, bolts, and adhesive for proper strength and balance.

The parabolic dish is then mounted on the top of the PVC shaft using a pulley and stud arrangement. The parabolic dish is made of satellite dish which is light in weight , and its surface is coated with reflective mirror pices to improve the reflection of sunlight. The shape of the dish is maintained accurately so that all reflected rays meet at one focal point. A copper tube painted in blackbody colour is fixed at this focal point using a supporting frame. This tube is connected with inlet and outlet pipes for water flow. To ensure smooth and stable rotation of the dish, a balancing weight is attached to the stud nut on the opposite side of the dish. This dead weight helps maintain proper balance of the parabolic reflector and prevents vibration or tilting during movement. The entire receiver assembly is held in place with metal nut & bolts , and insulation material is added back side the tube to prevent heat loss.

Next, the solar panel and battery system are installed on the same base frame. A 12V solar panel is fixed at a suitable angle to receive maximum sunlight. The panel output is connected to a 6V rechargeable battery through a charge controller. A PMDC motor is mounted on the base near to the shaft and connected to it using a V-belt and pulley arrangement of 25 cm diameter. Two LDR sensors are fixed on both sides of the parabolic dish, and all wiring between the panel, battery, sensors, and motor is neatly arranged using insulation tape and clips. After completing the assembly, the entire setup is checked for proper alignment, balance, and secure fitting of each component, ensuring the system is mechanically stable and ready for operation.

CHAPTER-6

PROGRAMING

```
# define east_button_pin 2
# define west_button_pin 3

int motr1 = 4;
int motr2 = 5; /

int eastLDR = 6;
int westLDR = 7;

int eastLDRreading = 0;
int westLDRreading = 0;

void setup(){
  pinMode(motr1, OUTPUT);
  pinMode(motr2, OUTPUT);

  pinMode(east_button_pin, INPUT_PULLUP);
  pinMode(west_button_pin, INPUT_PULLUP);

  pinMode(eastLDR, INPUT);
  pinMode(westLDR, INPUT);

  digitalWrite(motr1, LOW);
  digitalWrite(motr2, LOW);

  Serial.begin(9600);
  Serial.println("Design and Fabrication of Solar Air Heater With ");
  Serial.println("Active Solar Tracking");
}
```

```
void motorButtoneast(){  
  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, HIGH);  
    delay (2000);  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, LOW);  
    delay (1500);  
}
```

```
void motorButtonwest(){  
  
    digitalWrite(motr1, HIGH);  
    digitalWrite(motr2, LOW);  
    delay (2000);  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, LOW);  
    delay (1500);  
}
```

```
void motorLDReast(){  
  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, HIGH);  
    delay (2000);  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, LOW);  
    delay (1500);  
}
```

```
void motorLDRwest (){  
  
    digitalWrite(motr1, HIGH);  
    digitalWrite(motr2, LOW);  
    delay (2000);  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, LOW);  
    delay (1500);  
}
```

```
void motorStop (){  
  
    digitalWrite(motr1, LOW);  
    digitalWrite(motr2, LOW);  
}
```

```
void loop(){  
  
    byte east_button_state = digitalRead(east_button_pin);  
    byte west_button_state = digitalRead(west_button_pin);  
  
    eastLDRreading = digitalRead (eastLDR);  
    westLDRreading = digitalRead (westLDR);  
  
    Serial.print(" East = ");  
    Serial.println(eastLDRreading);  
    Serial.print(" West = ");  
    Serial.println(westLDRreading);  
    Serial.println("  ");
```

```
if (east_button_state == LOW){
    Serial.println("East Button Is Pressed");
    motorButtoneast();
}
else
    motorStop ();

if (west_button_state == LOW){
    Serial.println("West Button Is Pressed");
    motorButtonwest();
}
else
    motorStop();

if (eastLDRreading > westLDRreading){
    motorLDReast();
}
else
    motorStop();

if (eastLDRreading < westLDRreading){
    motorLDRwest();
}
else
    motorStop();

if (eastLDRreading == westLDRreading){
    motorStop();
}
}
```

CHAPTER-7

WORKING

The working of the Parabolic Solar Water Heater with Automatic Tracking and Power Generation is based on the principle of solar energy concentration and automatic solar tracking. The parabolic dish collects and focuses sunlight onto a single focal point. The surface of the dish is covered with mirror pieces, which act as reflective material to direct the sunlight accurately toward the center. A black-painted copper receiver tube is fixed at this focal point to absorb the maximum heat. When water passes through the receiver tube, it gains thermal energy from the concentrated solar rays, causing a steady rise in its temperature. The setup ensures that maximum solar radiation is utilized for heating the water efficiently.

The automatic solar tracking mechanism allows the parabolic dish to continuously face the sun from morning to evening. Two Light Dependent Resistors (LDRs) are mounted on opposite sides of the dish and connected to the control circuit. These sensors detect the difference in sunlight intensity between the two sides. When one sensor receives more light, the control circuit sends a signal to the microcontroller and with help of motor driver motor is start rotating, which rotates the dish slightly—about 4 to 5 degrees at a time—until both LDRs receive equal sunlight. This gradual movement keeps the dish properly aligned with the sun throughout the day, maintaining the correct focus on the receiver and improving overall heat absorption efficiency.

The electrical power system of the project operates through a 6V solar-powered battery. A solar panel converts sunlight into electrical energy and charges the 6V battery via a charge controller. The stored electrical energy is used to power the PMDC motor, LDR sensors, and the motor driver circuit for automatic tracking. A blocking diode is used in the circuit to allow one-way current flow, preventing reverse discharge of the battery at night. The combined operation of the solar panel, battery, sensors, and motor ensures that the system runs automatically without manual adjustment. Thus, the parabolic solar heater continuously tracks the sun, heats the water efficiently, and uses clean solar energy both for heating and for controlling its own movement.

CHAPTER-8

ADVANTAGES

1. Uses free and renewable solar energy.
2. Reduces electricity bills.
3. Eco-friendly and pollution-free operation.
4. Provides hot water without fossil fuels.
5. Easy to operate and maintain.
6. Increases efficiency through solar tracking.
7. Works even in remote areas without grid power.
8. Reduces dependency on conventional energy sources.
9. Compact and cost-effective design.
10. Provides both heating and power generation.
11. Promotes the use of clean energy technology.
12. Suitable for domestic and industrial use.
13. Helps conserve non-renewable energy resources.
14. Encourages awareness of solar energy applications.
15. Long service life with low running cost.

CHAPTER-9

APPLICATIONS

1. Domestic water heating for homes.
2. Solar cooking and food preparation.
3. Water heating for hostels, hospitals, and hotels.
4. Steam generation for small-scale industries.
5. Laundry and cleaning operations needing hot water.
6. Solar-powered water distillation systems.
7. Dairy processing (milk pasteurization, cleaning, etc.).
8. Power generation for small electrical appliances.
9. Agricultural drying and greenhouse heating.
10. Community solar kitchens and canteens.
11. Heating water for swimming pools.
12. Solar-based desalination plants.
13. Boiler feed water preheating in industries.
14. Educational and research purposes in renewable energy studies.
15. Emergency water heating in off-grid or remote locations.

CHAPTER-10

COST ESTIMATION

Sr. No.	Components	QYT	Materials	Cost (Rs)
1.	Parabola dish	1	Mild Steel	600
2.	Wooden Base	1	MDF Sheet	650
3.	Mirror Pieces	2500	Mirror	400
4.	Copper Tube	1	copper	450
5.	PVC Pipe & Iron Rod	1 (Each)	PVC & Iron	350
6.	End Caps	2	PVC	50
7.	Arduino Nano, Motor Driver, Push Buttons (2), LDR Sensors (2) & Wires	Set	-	1000
8.	PMDC Motor	1	-	150
9.	Battery	1	Lead Acid	400
10.	Stud Nut, Nut & Bolt	Set	Iron	150
11.	Dead Weight	1	Ceramic	50
12.	Pulley (25cm & 5cm) diameter	1 (Each)	Fiber and Brass	75
13.	Supply Pipes (5mm & 10mm)	1 (Each)	Polyethylene & Synthetic	100
14.	Paint and Insulation	Set	Body Paint & Aluminium Sheet	200

Sr. No.	Components	QYT	Materials	Cost (Rs)
15.	Special Glue	4	-	400
16.	Labour Cost	-	-	1000
17.	Auto	-	-	700
18.	Waste Material	-	-	275
	TOTAL		=	7000 RS

CHAPTER-11

CONCLUSION

The designed parabolic solar water heater successfully utilizes solar energy for water heating and small-scale power generation.

The use of a parabolic dish with mirror pieces ensures effective concentration of sunlight on the receiver tube.

The automatic single-axis tracking system increases the efficiency by keeping the dish aligned with the sun throughout the day.

A 6V DC motor controlled by LDR sensors enables gradual rotation of about 4–5 degrees per movement for accurate tracking.

The generated heat is stored in water, providing a renewable and eco-friendly source of energy. The attached solar panel also generates power to charge the 6V battery used for operating the tracking system.

This project demonstrates the practical use of solar energy in both heating and automation applications.

The system design is simple, cost-effective, and suitable for domestic and educational purposes. It helps in reducing dependence on conventional energy sources and promotes sustainable technology.

Overall, the project meets its objectives of energy conservation, automation, and efficient solar energy utilization.

CHAPTER-12

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